```
# Install required libraries
!pip install opencv-python matplotlib

import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# -----------------------------
# Upload Image
# -----------------------------
uploaded = files.upload()
image_path = list(uploaded.keys())[0]

# Read image
img = cv2.imread(image_path)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_RGB2GRAY)

# -----------------------------
# Define scales
# -----------------------------
scales = [0.5, 0.75, 1.0, 1.5, 2.0]

# ORB detector
orb = cv2.ORB_create(nfeatures=1000)

results = []
feature_counts = []

# -----------------------------
# Process each scale
# -----------------------------
for scale in scales:
    resized = cv2.resize(gray, None, fx=scale, fy=scale)

    keypoints, descriptors = orb.detectAndCompute(resized, None)

    img_kp = cv2.drawKeypoints(resized, keypoints, None, color=(0,255,0))

    results.append(img_kp)
    feature_counts.append(len(keypoints))

# -----------------------------
# Display Results
# -----------------------------
plt.figure(figsize=(15,6))

for i, scale in enumerate(scales):
    plt.subplot(1,5,i+1)
    plt.imshow(results[i], cmap='gray')
    plt.title(f"Scale {scale}\nFeatures: {feature_counts[i]}")
    plt.axis('off')

plt.tight_layout()
plt.show()

# -----------------------------
# Plot Feature Count vs Scale
# -----------------------------
plt.figure()
plt.plot(scales, feature_counts, marker='o')
plt.xlabel("Scale Factor")
plt.ylabel("Number of Features")
plt.title("Effect of Scaling on Feature Detection")
plt.grid()
plt.show()

# -----------------------------
# Print Results Table
# -----------------------------
print("------ FEATURE COUNT TABLE ------")
for i in range(len(scales)):
    print(f"Scale {scales[i]} -> Features Detected: {feature_counts[i]}")
```

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Requirement already satisfied: opencv-python in /usr/local/lib/python3.12/dist-packages (5.0.0.93)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: numpy>=2 in /usr/local/lib/python3.12/dist-packages (from opencv-python) (2.0.2)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.2)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.9.0.post
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib) (
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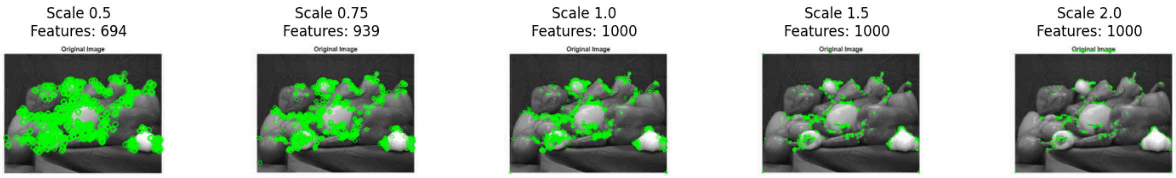
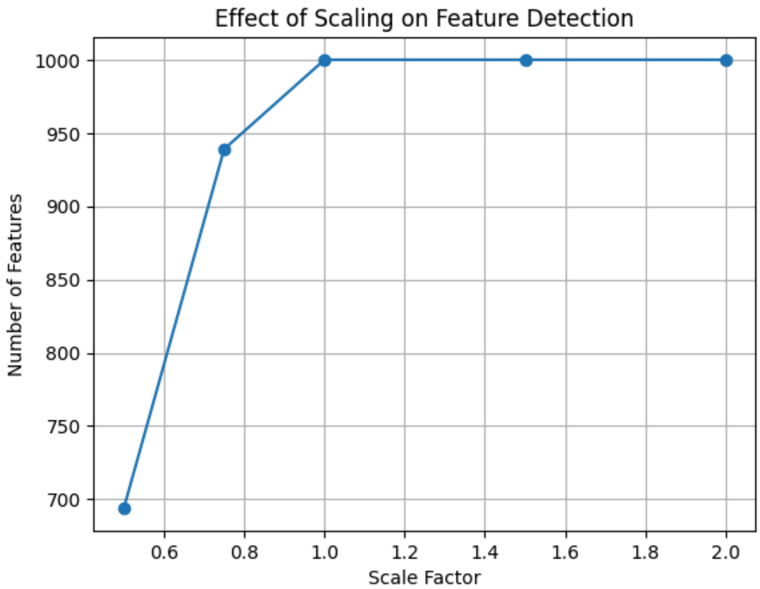
```






```
------ FEATURE COUNT TABLE ------
Scale 0.5 -> Features Detected: 694
Scale 0.75 -> Features Detected: 939
Scale 1.0 -> Features Detected: 1000
Scale 1.5 -> Features Detected: 1000
Scale 2.0 -> Features Detected: 1000
```